Taskflow — Apache JMeter Manual Testing

Assignment Submission Report

Project: Taskflow — Daily Task / Check-in Application

Testing Tool: Apache JMeter 5.6.3

Test Environment: Localhost demo API (127.0.0.1:8000), FastAPI/Uvicorn

This report documents a manual functional testing approach for the Taskflow project using Apache JMeter. Because the production Taskflow backend was not yet available, a lightweight local API was created solely to provide stable endpoints for demonstrating the testing workflow.

1. Objective

To explore manual testing in Apache JMeter and apply it to the ongoing Taskflow project by designing functional test cases, executing HTTP requests, observing responses, and recording the results.

2. Test Environment

Component	Configuration
Operating system	Windows
JMeter	Apache JMeter 5.6.3
Application	Taskflow
Backend	FastAPI + Uvicorn
Base URL	http://127.0.0.1:8000
Test users	1
Loop count	1

3. Test Strategy

The test plan uses a single JMeter Thread Group with one virtual user and one iteration. The cases cover basic availability, CRUD operations, and a negative test. The CRUD sequence creates a task, captures its ID, retrieves it, updates it, and finally deletes it.

4. Test Cases and Results

ID	Test	Method	Expected	Observed	Status
TC-01	Health Check	GET	200	200	PASS
TC-02	Open Taskflow	GET	200	200	PASS
TC-03	List Tasks	GET	200	200	PASS
TC-04	Create Task	POST	201	201	PASS
TC-05	Get Created Task	GET	200	200	PASS
TC-06	Update Created Task	PUT	200	200	PASS
TC-07	Delete Created Task	DELETE	200	200	PASS
TC-08	Invalid Task	GET	404	404 expected	PASS

5. Execution Evidence

An earlier successful JMeter execution of the same functional flow produced 8 samples. TC-01 through TC-07 returned successful HTTP responses. TC-08 returned HTTP 404 as designed for a non-existent task. JMeter initially reported this expected 404 as an error, producing a raw error rate of 12.50%; the final test plan normalizes the expected 404 to PASS.

Important: The 12.50% value is a raw JMeter result from the initial execution before the expected-404 handling was added. The test case itself is not a defect because 404 is the specified expected result for TC-08.

6. Defects / Observations

Observation	Assessment
Health endpoint responded successfully	PASS
Task CRUD flow completed successfully	PASS
Invalid task returned 404	Expected negative behavior
Raw JMeter error count included the expected 404	Resolved in final plan using expected-response handling
Production backend unavailable	Local FastAPI demo backend used for assignment demonstration

7. Conclusion

The Taskflow application workflow was successfully represented as a manual functional test plan in Apache JMeter. The test suite covers availability, task creation, retrieval, update, deletion, and invalid-resource handling. The resulting JMX file, test-case matrix, backend demonstrator, and this report provide the material required to demonstrate the assignment work.

8. Submission Contents

• Taskflow_JMeter_FINAL.jmx — JMeter test plan

• Taskflow_Test_Cases.csv — test case matrix

• Taskflow_JMeter_Assignment_Final_Report.pdf — this report

• app.py — local FastAPI backend used for the demonstration

• requirements.txt — Python dependencies

• run_taskflow.bat — Windows launcher for the backend

9. JMeter Configuration Evidence

The following screenshots document the configured JMeter test plan and the execution/debugging state observed during preparation. The final test-case matrix and report above should be used as the formal submission evidence.



